Characterization and evidence: Worked solutions

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

E01 Time resolution

Frames 1 through 4 occur at 0, 30, 60 and 90 ms. The stated one-time detectable transition is in $(60, 90]$ ms. This bounds **detectable appearance**, conditional on instantaneous sampling and persistence. Finite exposure, clock synchronization error, transient visibility or a detection threshold change or complicate the bound.

E02 Tunneling gap

The unchanged factors cancel: $I_2/I_1 = e^{-2\kappa\Delta z} = e^{-2} \approx 0.135$. In constant-current imaging, feedback changes tip height to maintain the setpoint, so local electronic states, tip states and bias can affect recorded height as well as actual geometry.

E03 Gaussian width

Gaussian variances add. Since FWHM is proportional to standard deviation, $w_{obs}^2 = w_{sample}^2 + w_{inst}^2$, giving $w_{sample} = \sqrt{10^2 - 6^2} = 8$ nm. A nonlinear or changing tip contact, or non-Gaussian signal/response, invalidates this simple model. A conducting profile can also differ from the geometric filament cross-section.

E04 Slice interpolation

Between 0 and 2 nm, $I(1) = 2 + (1/2)(6 - 2) = 4$ nA; between 2 and 4 nm, $I(3) = 6 + (1/2)(10 - 6) = 8$ nA. On the second interval $I(z) = 6 + 2(z - 2)$ nA with z in nm, so 7 nA occurs at $z = 2.5$ nm. These are model estimates, not measured additional slices.

E05 Pressure modulation

$R_0 = 1/(10\,\mu\text{S}) = 100\,\text{k}\Omega$; $R_1 = 1/(12\,\mu\text{S}) \approx 83.33\,\text{k}\Omega$. At $1\,\mu$A, $V_0 = 0.100$ V and $V_1 \approx 0.0833$ V. The fractional conductance change is $(12 - 10)/10 = 0.20 = 20\%$. The voltage dip indicates a resistance decrease in this fixed-current example, not a universal response: sign depends on material, state, force and mechanism.

E06 Measurement design

Compare low-bias scans and an unprogrammed control to test read disturbance or drift; scan a reference surface or replace the tip to assess wear and contact change; correlate low-disturb maps with repeated SET/RESET and composition-sensitive imaging. A hotspot alone cannot identify a chemical species or establish a geometric filament diameter. Each control has its own perturbations and resolution limit.

E07 In situ imaging and chemical evidence

Brightness indicates contrast under particular imaging conditions and can reflect thickness, geometry, orientation or composition. Synchronized SET/RESET current and images test whether the feature follows the electrical state; spatially resolved EDS/EELS with suitable background and standards tests elemental or chemical-state hypotheses. Check projection overlap, thinning, beam dose, vacuum and whether the device continues to switch; correlation alone does not prove a unique mechanism.

Characterization and evidence: Worked solutions (continued)

E08 Reconstruction limits

Depth calibration assigns each slice a z position; registration aligns lateral features; interpolation fills unmeasured locations using a model; threshold selects which reconstructed values form the displayed isosurface. Each can change apparent shape. Sequential removal is destructive and takes time, so slices do not form an instantaneous view of an unaltered device. A reference electrode or stable reference region can help track probe changes between slices.

E09 Unresolved feature

Nominal subtraction is $5.8^2 - 6.0^2 < 0$, so no positive real width follows from the nominal Gaussian model. Uncertainty intervals overlap, so do not report a negative width or force it to zero as an exact diameter. State that the feature is unresolved and, if justified, calculate a model-based upper limit using an explicitly specified uncertainty method.

E10 Compare CAFM configurations

The first experiment writes with intact device electrodes, then exposes the layer; the second uses a movable tip as electrode during writing and reading. Contact area, field, force and environment differ. Electrode removal can damage or oxidize a channel; a tip can wear, contaminate or electrically disturb a film. Different maps can reflect changed boundaries or preparation, not necessarily contradictory intrinsic behavior.

E11 Retention or scan artifact?

Record a low-disturb baseline for the target and an unprogrammed region. Apply a documented pulse only to the target, then image both immediately and after a fixed wait with the same read protocol. Check tip condition against a reference surface or with a fresh tip and compare topography/current for drift. A persistent target-specific contrast supports a retained local electrical response under those conditions. It does not by itself prove a metal filament, composition or exact dimensions.

E12 Coherent double barrier

Initially, both phases give $\sin^2(\phi/2) = 1/2$ and $T = 1/(1 + 3/2) = 2/5$. From $\phi = \pi/2$, increasing phase to $2\pi/3$ gives $T = 1/(1 + 3 \cdot 3/4) = 4/13$, hence $\Delta G/G = (4/13 - 2/5)/(2/5) = -3/13 \approx -23.1\%$. From $\phi = 3\pi/2$, increasing phase to $5\pi/3$ gives $T = 1/(1 + 3 \cdot 1/4) = 4/7$, hence $\Delta G/G = (4/7 - 2/5)/(2/5) = 3/7 \approx +42.9\%$. Opposite signs arise from the assumed phase positions; a pressure-induced positive phase shift is not universal. Dephasing, finite-temperature averaging or multiple channels with differing phases may weaken or alter interference; pressure may also change the barriers.

Characterization and evidence: Worked solutions (continued)

E13 EDS evidence and limitations

Subtracting the stated background gives $80 - 20 = 60$ net counts after SET and $50 - 20 = 30$ after RESET, so the net-count ratio is $60/30 = 2$. This is an invented counting example. Equal acquisition exposure and collection geometry are not enough to establish a concentration ratio: thickness, detector response, spectral overlap and counting uncertainty can also change it. Align the spectrum acquisition interval with the voltage/current record and image frame intervals; use a lower-dose or unbiased control and compare whether the specimen still switches after preparation and imaging. A state-correlated elemental signal supports a chemical hypothesis but need not exclude every competing mechanism.

E14 Thresholded slice area

Each cell represents $10 \times 10 = 100\,\mathrm{nm}^2$. For $I \geq 4$ nA the selected values are 4, 8 and 6 nA, so the displayed area is $3 \times 100 = 300\,\mathrm{nm}^2$. For $I \geq 7$ nA only 8 nA remains, giving $100\,\mathrm{nm}^2$. The calculation changes the displayed threshold, not the physical specimen: contact response, current sensitivity and the chosen threshold mean neither area is necessarily the true filament cross-section. Registration is needed to compare different slices; tracking a reference or tip calibration helps reveal changes in contact response between them.

E15 Growth model and state-dependent evidence

Under the stated qualitative model, high ion mobility favors transport to the inert electrode before nucleation; slow redox can favor branching at high-field filament tips. This is a conditional prediction, not a conclusion about every specimen with those labels. For the Pt/TiO_2/Pt case, a resistance change correlated with the loss of a Ti_4O_7-related diffraction reflection supports a structural contribution. Check the same specimen and region before/after RESET and the diffraction orientation and acquisition conditions; projection, thickness or orientation changes can affect the observed reflection, and chemical changes may accompany it. For the separate Ta-oxide case, state-dependent STEM-EELS spectra support a difference in local electronic or chemical state. The authors interpreted their specimen's LRS as metallic TaO and HRS as an insulating mixed-valence Ta_2O_5–TaO_2 region. Compare calibrated reference spectra, background, beam dose and registered regions; different thickness or beam-induced changes are alternative influences. Neither image contrast nor an isolated spectral change uniquely establishes a phase or the complete conducting path.